JAVA PROJECT DOCUMENTATION
Hospital Registration System

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Hospital registration system (Java + MySQL + GUI) Documentation

1. Project Overview

The Hospital registration system is a Java-based desktop application integrated with MySQL using JDBC for backend data management. It provides two primary user roles: **Patient** and **Doctor**. Patients can register themselves and choose from available dentists, while doctors can manage appointments.

2. Technologies Used

Programming Language: Java

Database: MySQL (XAMPP - MariaDB)

GUI Framework: Java Swing

Database Connectivity: JDBC

3. Features

Patient Role:

Registration with personal details: name, age, gender, contact

Dropdown to choose available dentists

Stores selected doctor into database along with patient info

Doctor Role:

View list of dentists and choose their name to make appointment for their patient

4. Database Schema

Database: hospital_db

Tables:

`doctor`:

`doctor_id` INT PRIMARY KEY AUTO_INCREMENT

`name` VARCHAR(100)

`specialization` VARCHAR(50)

`availability` BOOLEAN

`patient`:

- `patient_id` INT PRIMARY KEY AUTO_INCREMENT
- `name` VARCHAR(100)
- `age` INT
- `gender` VARCHAR(10)
- `contact` VARCHAR(20)
- `selectedDoctor` VARCHAR(50) (stores doctor name or ID)

`appointment`:

- `appointment_id` INT PRIMARY KEY AUTO_INCREMENT
- `patient_id` INT (FK to patient)
- `doctor_id` INT (FK to doctor)
- `appointment_date` DATE
- `status` VARCHAR(20)

5. Key Java Classes

entity.Doctor.java

Displays available dentists in JComboBox

On selection, passes doctor ID to Appointment window

entity.Patient.java

GUI to register patient details and choose doctor

Passes data to `dao.PatientDAOo`

dao.PatientDAOo.java

Inserts patient data into `patient` table using `PreparedStatement`

entity.Appointment.java

Accepts doctor ID and provides appointment interface (can be extended for real booking)

db.DBconection.java

Manages DB connection using JDBC

6. Sample Data

```
INSERT INTO doctor (name, specialization, availability) VALUES
('Dr. Rediet', 'Dentist', TRUE),
('Dr. Natnael', 'Dentist', TRUE);
```

7. Common Errors & Fixes

Ensure `Appointment` constructor exists and matches call signature

Database columns must match insert statement fields

Use correct JDBC URL: `jdbc:mysql://localhost:3306/hospital_db`

8. Future Improvements

Extend Appointment module to select date/time

Add authentication for doctors and patients

Include medical history and prescriptions tables

Add real-time validation and better UI design

9. How to Run

Start XAMPP and ensure MySQL is running

Import SQL schema into phpMyAdmin

Compile and run the Java application

Register as Patient or log in as Doctor